

D179

Data-Informed Practices

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A teacher with a well-organized grade book asks 28 students, in a survey on a new tablet app, to “Describe one thing that made today's lesson hard.” The survey took five minutes. What type of data do the answers produce?

- A. Qualitative data.
- B. Demographic data.
- C. Quantitative data, because 28 students answered.
- D. Quantitative data, because it came from a survey.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A school team, meeting in a newly remodeled conference room, reviews a spring survey in which 64% of families said they feel welcome at school. The survey was translated into three languages. What type of school data is this?

- A. Demographic data.
- B. School process data.
- C. Student learning data.
- D. Perceptual data.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A team wants to know whether its new tutoring schedule is actually being followed. Which type of data should it collect?

- A. Demographic data on tutored students.
- B. School process data, such as tutoring logs.
- C. Perceptual data, such as a student opinion poll.
- D. Student learning data, such as test scores.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A curriculum coach who has worked in the district for twelve years downloads last year's state test results for her 300 students from the state website. What kind of data is this for her question about her own students?

- A. Secondary data.
- B. Qualitative data.
- C. Primary data, because she downloaded it herself.
- D. Primary data, because it is about her students.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
-----------------------------	---	-----------------------	----------------------------------

5.

A vendor's glossy brochure, printed on heavy paper, says its reading app raised scores 40% in "partner schools." No sample size or method is given, and the study was paid for by the vendor. The app has a friendly mascot. How should a teacher treat this claim?

- A. As useless, since vendor information is always false.
- B. As strong evidence because 40% is a large gain.
- C. As weak evidence until an independent study with a clear method and sample confirms it.
- D. As strong evidence, since vendors know their products best.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
-----------------------------	---	-----------------------	----------------------------------

6.

Seven students logged weekly reading minutes on a new app: 20, 22, 25, 25, 28, 30, and 180. The student with 180 read during a family road trip. The teacher wants one number that shows typical reading time. Which should she report?

- A. 180 minutes.
- B. The mean without any value above 25, about 23 minutes.
- C. The mean, about 47 minutes.
- D. The median, 25 minutes.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
-----------------------------	---	-----------------------	----------------------------------

7. [TRANSFER — reversed stem / different variable]

A report on family commute times shows a mean of 62 minutes and a median of 40 minutes. What does this suggest about the data?

- A.** A few very short commutes are pulling the mean up.
- B.** A few very long commutes are pulling the mean up.
- C.** The data contain an error, since mean and median must match.
- D.** Most families commute about 62 minutes.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A school where most teachers have ten or more years of experience reports that 200 of its 250 fifth graders (80%) passed the math test. Of those, 200 students are in general classes and 180 of them passed; the other 50 are English learners. What is the pass rate for English learners?

- A.** 10%
- B.** 40%
- C.** 90%
- D.** 80%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A principal says, “Our overall graduation rate is 91%, so every group is doing fine.” What is the best response?

- A.** Report the 9% who did not graduate as one group.
- B.** Compare 91% with last year's overall rate.
- C.** Break the rate out by subgroup before drawing that conclusion.
- D.** Agree, since 91% is above the state target.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Ninth-grade failure rates are high at a school with a new football stadium. A team finds that many failing students missed over 15 days, and asking why shows first-period absences spike because the bus arrives after the bell. Which is the root cause the school can act on?

- A.** High ninth-grade failure rates.
- B.** Spending on the stadium.
- C.** The bus schedule brings students after first period starts.
- D.** Students missing more than 15 days.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
-----------------------------	---	-----------------------	----------------------------------

11.

A teacher who runs a popular chess club wants to try exit tickets to improve daily feedback. The department of nine teachers is interested. Using a Plan-Do-Study-Act approach, what should happen first?

- A.** Have all nine teachers use them for the full year.
- B.** Start with a survey of the chess club.
- C.** Adopt them without data, since feedback is known to help.
- D.** Test exit tickets in one class for two weeks, collect data, and then decide whether to adopt, adapt, or drop them.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
-----------------------------	---	-----------------------	----------------------------------

12. [TRANSFER — reversed stem / different variable]

After a two-week test, data show a new warm-up routine did not improve on-time starts. In a PDSA cycle, what comes next?

- A.** Adapt the routine or drop it, then plan another small test.
- B.** Stop using data for this problem.
- C.** Repeat the same test without changes.
- D.** Roll it out schoolwide anyway.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
-----------------------------	---	-----------------------	----------------------------------

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A teacher ~~with a well-organized grade book~~ asks 28 students, in a survey on a new tablet app, to “Describe one thing that made today's lesson hard.” ~~The survey took five minutes.~~ What type of data do the answers produce?

Target word: data type **Decides:** The answers are open-ended descriptions.

A	Correct	Written descriptions of experience are qualitative.
B	Adjacent category	Demographic data describe who students are, not their views.
C	Wrong stem number	The number of responders does not make the answers numeric.
D	Confusion set	The tool does not set the type; the form of the answers does.

2. Answer D Concept 2

A school team, ~~meeting in a newly remodeled conference room,~~ reviews a spring survey in which 64% of families said they feel welcome at school. ~~The survey was translated into three languages.~~ What type of school data is this?

Target word: classify the data **Decides:** Families' feelings and opinions.

A	Camouflage	The language detail describes the survey, not what it measured.
B	Confusion set	Process data describe programs and practices.
C	Confusion set	Learning data measure achievement.
D	Correct	Opinions and feelings are perceptual data.

3. Answer B Concept 2 · transfer

A team wants to know whether its new tutoring schedule is actually being followed. Which type of data should it collect?

Target word: choose type from the question **Decides:** The question is about a practice being carried out.

A	Dropped element	Describes who, not whether the schedule ran.
B	Correct	Process data describe how programs run.
C	Confusion set	Opinions do not confirm what happened.
D	Adjacent category	Scores show results, not whether the schedule was followed.

4. Answer A Concept 3

A curriculum coach ~~who has worked in the district for twelve years~~ downloads last year's state test results for her 300 students from the state website. What kind of data is this for her question about her own students?

Target word: primary or secondary **Decides:** The state collected it for its own purpose.

A	Correct	Existing data collected by someone else for another purpose.
B	Adjacent category	Answers a different question: form, not origin.
C	Confusion set	Retrieving data is not collecting it.
D	Inverted operation	Primary depends on who collected it, not whom it describes.

5. Answer C Concept 4

A vendor's glossy brochure, ~~printed on heavy paper~~, says its reading app raised scores 40% in "partner schools." No sample size or method is given, and the study was paid for by the vendor. ~~The app has a friendly mascot.~~ How should a teacher treat this claim?

Target word: credibility judgment **Decides:** No method, no sample, and a funder with a stake.

A	Absolute language	Vendor claims need checking, not automatic rejection.
B	Wrong stem number	The size of a claim does not make it credible.
C	Correct	Missing method and a conflict of interest lower credibility.
D	Inverted operation	A stake in the outcome lowers independence.

6. Answer D Concept 5

Seven students logged weekly reading minutes on a new app: 20, 22, 25, 25, 28, 30, and 180. ~~The student with 180 read during a family road trip.~~ The teacher wants one number that shows typical reading time. Which should she report?

Target word: typical value with an outlier **Decides:** 180 is far above the rest; median resists it.

A	Wrong stem number	The outlier, not the typical value.
B	Dropped element	Throws out valid data rather than choosing a resistant measure.
C	Adjacent category	$330 / 7 \approx 47$, pulled up by one extreme value.
D	Correct	Middle of seven sorted values is 25, unaffected by 180.

7. Answer B Concept 5 • transfer

A report on family commute times shows a mean of 62 minutes and a median of 40 minutes. What does this suggest about the data?

Target word: read skew from mean vs. median **Decides:** Mean well above median.

A	Inverted operation	Low values would pull the mean down.
B	Correct	High outliers raise the mean but not the median.
C	Absolute language	They differ whenever data are skewed.
D	Confusion set	The median, 40, is the better typical value here.

8. Answer B Concept 6

A school ~~where most teachers have ten or more years of experience~~ reports that 200 of its 250 fifth graders (80%) passed the math test. Of those, 200 students are in general classes and 180 of them passed; the other 50 are English learners. What is the pass rate for English learners?

Target word: subgroup rate **Decides:** EL passers = $200 - 180 = 20$ of 50.

A	Inverted operation	Divides 20 by 200 instead of by the 50 English learners.
B	Correct	$20 / 50 = 40\%$, far below the 80% overall.
C	Confusion set	That is the general-class rate, $180 / 200$.
D	Dropped element	The overall rate hides the subgroup gap.

9. Answer C Concept 6 • transfer

A principal says, “Our overall graduation rate is 91%, so every group is doing fine.” What is the best response?

Target word: flaw in reasoning from an overall rate **Decides:** An average can hide subgroup gaps.

A	Dropped element	Still does not look at subgroups.
B	Adjacent category	A trend of the same average still hides subgroups.
C	Correct	Disaggregation shows whether gaps exist.
D	Absolute language	A high overall rate does not guarantee each group is high.

10. Answer C Concept 7

Ninth-grade failure rates are high at a school ~~with a new football stadium~~. A team finds that many failing students missed over 15 days, and asking why shows first-period absences spike because the bus arrives after the bell. Which is the root cause the school can act on?

Target word: deepest actionable cause **Decides:** Bus arrives after the bell.

A	Inverted operation	That is the problem, not its cause.
B	Camouflage	Nothing in the data links the stadium to failure.
C	Correct	The deepest cause found, and the school can act on it.
D	Dropped element	A symptom; the team found a deeper why.

11. Answer D Concept 8

A teacher ~~who runs a popular chess club~~ wants to try exit tickets to improve daily feedback. The department of nine teachers is interested. Using a Plan-Do-Study-Act approach, what should happen first?

Target word: how to start a change **Decides:** PDSA tests small first.

A	Absolute language	Scales before testing.
B	Camouflage	The club is unrelated to the change.
C	Dropped element	Skips the Study step.
D	Correct	A small test with data comes before scaling.

12. Answer A Concept 8 • transfer

After a two-week test, data show a new warm-up routine did not improve on-time starts. In a PDSA cycle, what comes next?

Target word: the Act step after a failed test **Decides:** Study showed no gain; Act means adapt or abandon.

A	Correct	Act uses the results to change course and cycle again.
B	Absolute language	One failed test is part of the cycle, not the end of it.
C	Adjacent category	Rerunning without adapting ignores what was learned.
D	Inverted operation	Scales a change the data did not support.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Quantitative vs. qualitative data	Quantitative data are numbers that can be counted or measured; qualitative data are words, images, or observations that describe qualities and reasons.	Calling survey data quantitative just because it came from a survey, even when answers are open-ended comments.	The form the data take: numbers or descriptions.
2	Four types of school data	Demographic data describe who students are; perceptual data capture opinions and beliefs; student learning data show achievement; school process data describe programs and practices.	Classifying a satisfaction survey as learning data.	A data source to be labeled by type.
3	Primary vs. secondary data	Primary data are collected by you for your question; secondary data already exist, collected by someone else for another purpose.	Calling state test data primary because it is about your own students.	Who collected the data and for what purpose.
4	Credibility of a data source	Judge a source by its method, sample, recency, and independence from those who benefit from the result.	Trusting a polished vendor report because it has charts and big numbers.	A claim from a source with a stake in the outcome.
5	Mean vs. median with outliers	An extreme value pulls the mean toward it; the median better shows the typical value when data are skewed.	Reporting the mean as typical when one value is far from the rest.	A small data set with one extreme number.
6	Disaggregation	Break results out by subgroup, because an overall average can hide large gaps between groups.	Concluding everyone is doing well because the overall rate looks strong.	A single overall percentage and a question about who is struggling.
7	Root cause vs. symptom	Ask why repeatedly until reaching a cause the school can act on; treating a symptom leaves the problem in place.	Stopping at the first visible problem, such as low scores, and calling it the cause.	A problem statement and a choice among causes at different depths.
8	Plan-Do-Study-Act cycles	Continuous improvement tests a small change, studies data on its effect, and then adopts, adapts, or abandons it before scaling.	Rolling out a big change everywhere at once with no test.	A proposed change and a question about how to try it.

Course scope (public description): Data-Informed Practices focuses on the development of data literacy skills. This course teaches candidates about the different types of data, the benefits and limitations of those data types, and how they can use data to identify and solve problems and inform decisions. The course also teaches candidates how to locate, collect, and analyze data from relevant and credible sources, and how to draw conclusions from data in order to drive continuous improvement. There are no prerequisites for this course.